* Important Array interview questions: → { Arrays, Strings }

Jagged Array → 2D Array with unequal rows

{ { 1, 2, 3 }, → r(0) = 3

Jagged { 2, 4, 8, 9 }, r(1) = 4

Array { 1, 2 } r(2) = 2 }

* Prefix Sum: →

Sum Till index:

arr = { 1, 2, 3, 4 }
0 1 2 3

{ 1, 3, 6, 10 }

prefix [4] → p[0] = a[0] o/p ⇒ { 1, 3, 6, 10 }

1 2 3

p[i] = p[i-1] + a[i]

p[1] = p[0] + a[1]
= 1 + 2 = 3

p[2] = p[1] + a[2]
= 3 + 3 = 6

p[3] = p[2] + a[3]
= 6 + 4
= 10

0 0 0 0 0
0 1 2 3 4

add 10 to the elements from index 1 to 3

o/p 0 10 10 10 0

Difference Sum (1-3)

1, 2, 3, 4, 5

1, 7, 8, 9, 5

5

box0	b1	b2	b3	b4
5	8	9	2	7
c	c	c	c	c
	+	+	+	
	3	3	3	

1st – 3rd

5 11 12 5 7

n = 5 diff[] = int[5] 0, 0, 0, 0, 0
0 1 2 3 4
l r

int l = 1
int r = 3
int val = 10;
diff[l] += val;
0 += 10

0 10 0 0 0

wrong ⇐ 0 10 10 10 10 ✗

0 10 0 0 -10
0 10 10 10 0

if (r+1) < n ⇒ diff[r+1] -= val;
10
4 < 5

diff[4] = -10;

normal prefix sum

* Majority Element: → Boyer Moore's algo for Voting

int[] arr = { 2, 2, 1, 1, 2, 2, 2 };
0 1 2 3 4 5 6

count = 0;
candidate = 0;

for (0 → 6) {
if (count == 0) {
candidate = arr[i]
}
if (candidate == arr[i])
count ++
else count --
}

f( )
O(n)

Count	candidate	index
1	2	0
		1
2	2	2
1	2	3
0	2	4
1	2	5
2	2	6
3	2	

(Sparse Matrix) (No of zeros > No of non zeros)

0	0	3
0	0	0
0	1	2

Sparse

1	2	3
0	1	2
0	1	5

Not Sparse

(TVR) ⇒ count > $\frac{r \times c}{2}$

2D (2 > 4) $\frac{9}{2}$ = 4

1st 6 > 4

Array (Partitioning): → 3, 8, 6, 12, 7, 6

Partition the array in such a way that all even numbers are to the left & all odd numbers are to the right.

(Two Pointer) o/p: 6, 8, 12, 5, 7, 3

left = 0 right = n-1;
while (l < right)
while (arr[l] is even) l++
while (arr[r] is odd) r--
if not: (swap)